Matias Tupinamba

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Education

Texas A&M University Aug. 2025 – May 2029
Bachelor of Science in Electrical Engineering, Minor in Mathematics *College Station, TX*
Relevant Coursework: Digital Logic Design, Machine Learning, Linear Algebra, Physics E&M *GPA: 4.0*

Experience & Leadership

Jones STEM Club Representative June 2026 – Present
Texas A&M University — Society of Hispanic Professional Engineers (SHPE) *College Station, TX*

- Organize and execute elementary school volunteering program, creating 20+ biweekly activities that translate STEM principles into engaging experiences for classes of 15+ Jones Elementary School students
- Coordinate event logistics between TAMU SHPE and Jones Elementary School faculty to ensure well-run program
- Foster relationships with local STEM educators and K-12 students while creating curriculum

Lead Coding Instructor June 2026 – Aug. 2026
iCode STEAM Academy *Katy, TX*

- Formulated and instructed an AI engineering curriculum covering machine learning, APIs, and generative AI
- Developed full-stack web applications with 30+ students to provide hands-on experience with AI and programming
- Instructed 30+ students across classes of 8–12 on core Python and JavaScript concepts, facilitating the completion of 15+ student-led portfolio projects

Projects

Traffic Light Controller | *Verilog, FPGA, Digital Logic Design, Vivado* Oct. 2026

- Designed and verified a finite-state machine traffic light controller in Verilog, writing testbenches to simulate state transitions, sensor inputs, and variable timing intervals across multi-way intersections
- Deployed car sensor inputs to trigger green lights and adjust traffic flow based on real-time vehicle demand
- Tested and verified the system on a Zybo FPGA board using Vivado to map signal outputs to onboard LEDs

Mini Haptic Steering Wheel Controller | *Python, C++, Arduino, CAD* Jan. 2026

- Won best beginner hardware hack at TAMUHACK2026, a software & hardware hackathon with 800+ attendees
- Built custom force-feedback system using Moteus BLDC controller, transforming live data into physical feedback
- Implemented cross device communication between racing simulator, steering wheel, and force feedback device
- Utilized Fusion360 to construct the controller's base and housing to ensure compliance within size constraints

SPORA, SHPEATHON | *Project Management, Product Design* Jan. 2026

- Led team of 4 in engineering design competition from concept to final presentation, resulting in a top 5 placement
- Spearheaded design and prototyping of an eco-friendly mycelium erosion-control mat to replace petroleum-based geotextiles and curb environmental impact of construction
- Served as project's CEO and COO, aligning team responsibilities and keeping the project on schedule
- Developed and collaborated on a project plan which included product design, cost estimates, and presentation

Eagle Scout Project, Scouting America | *CAD, Project Management* Aug. 2024

- Directed a team of 10+ volunteers across 20+ service hours, enforcing field safety protocols and delegating tasks
- Modeled and executed the design, budgeting, and fabrication of custom bat habitats for Fort Bend County Parks & Recreation, utilizing CAD software to produce durable, weather-resistant structures intended to last 5+ years
- Managed a $250+ project budget by estimating material quantities, comparing vendor costs, and sourcing weather-grade cedar and fasteners to maximize long-term structural durability

Technical Skills

Programming: Python, C++, Verilog, JavaScript, HTML/CSS
Technical: Arduino, FPGA, 3D Modeling & Printing, Blender, CAD, Microsoft Office
Languages: Fluent in English & Portuguese, Intermediate Spanish